Lab2:

Checkpoint 1 — Decrypt Caesar Cipher

Caesar Cipher?

It's one of the simplest encryption methods.
Each letter is shifted by a fixed number of positions in the alphabet.

We try all possible shifts (1 to 25) and here one gives readable English.

Shift 0: odroboewscdrolocdcwkbdmyxdbkmdzvkdpybwyeddrobo

Shift 1: ncqnandvrbcqnknbcbvjaclxwcajlcyujcoxavxdccqnan

Shift 2: mbpmzmcuqabpmjmabauizbkwvbzikbxtibnwzuwcbbpmzm

Shift 3: laolylbtpzaolilzazthyajvuayhjawshamvytvbaaolyl

Shift 4: kznkxkasoyznkhkyzysgxziutzxgizvrgzluxsuazznkxk

Shift 5: jymjwjzrnxymjgjxyxrfwyhtsywfhyuqfyktwrtzyymjwj

Shift 6: ixliviyqmwxlifiwxwqevxgsrxvegxtpexjsvqsyxxlivi

Shift 7: hwkhuhxplvwkhehvwvpduwfrqwudfwsodwiruprxwwkhuh

Shift 8: gvjgtgwokuvjgdguvuoctveqpvtcevrncvhqtoqwvvjgtg

Shift 9: fuifsfvnjtuifcftutnbsudpousbduqmbugpsnpvuuifsf

Shift 10: ethereumisthebestsmartcontractplatformoutthere

Shift 11: dsgdqdtlhrsgdadrsrlzqsbnmsqzbsokzsenqlntssgdqd

Shift 12: crfcpcskgqrfczcqrqkypramlrpyarnjyrdmpkmsrrfcpc

Shift 13: bqebobrjfpqebybpqpjxoqzlkqoxzqmixqclojlrqqebob

Shift 14: apdanaqieopdaxaopoiwnpykjpnwyplhwpbknikqppdana

Shift 15: zoczmzphdnoczwznonhvmoxjiomvxokgvoajmhjpooczmz

Shift 16: ynbylyogcmnbyvymnmgulnwihnluwnjfunzilgionnbyly

Shift 17: xmaxkxnfblmaxuxlmlftkmvhgmktvmietmyhkfhnmmaxkx

Shift 18: wlzwjwmeaklzwtwklkesjlugfljsulhdslxgjegmllzwjw

Shift 19: vkyvivldzjkyvsvjkjdriktfekirtkgcrkwfidflkkyviv

Shift 20: ujxuhukcyijxuruijicqhjsedjhqsjfbqjvehcekjjxuhu

Shift 21: tiwtgtjbxhiwtqthihbpgirdcigprieapiudgbdjiiwtgt

Shift 22: shvsfsiawghvspsghgaofhqcbhfoqhdzohtcfacihhvsfs

Shift 23: rgurerhzvfgurorfgfznegpbagenpgcyngsbezbhggurer

Shift 24: qftqdqgyueftqnqefeymdfoazfdmofbxmfradyagfftqdq

Shift 25: pespcpfxtdespmpdedxlcenzyeclneawleqzcxzfeespcp

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Shift 10: ethereumisthebestsmartcontractplatformoutthere

Checkpoint 2 — Breaking Two Substitution Ciphers

What is a substitution cipher?

Unlike Caesar (which shifts all letters equally), a **substitution cipher** replaces each letter with another **random letter**.

Which ciphertext is easier to break, and why?

Breaking Ciphertext 1

Score (lower = better): 10.04

INOGORAICULORONDINEOCHCOSEDIMMERENAFOYAHESEMTURFEREINDISGENS
OBLEATHIWYUPTOWORYLBECOUSETMHISKUICVUNDERSAONDINPTMAHEGRINC
IGLESTMGSYCHTHISATRYONDTMHISIWOPINOAIJEGRTBINPSINATNEFOREOSIAF
OSCTWMTRAINPATVNTFAHOAIMONYAHINPHOGGENEDATSELDTNHIWSELMBEM
TREAHEWOAHEWOAICSTMAHEMIELDCTULDBECTWGLEAELYFTRVEDTUAONDH
TFSLTFLYIAGRTCEEDEDONDHTFWTUNAOINTUSAHETBSAOCLESAHEREFTULDO
ALEOSAREWOINTNEPTTDWINDAHOAFTULDCTNAINUEAHERESEORCH ...

Breaking Ciphertext 2

Score (lower = better): 17.90

CILCOGAHVERBRIYNASDVERBPEYULIARASDNADCEESTNEGOSDEROMTNEHNIR
EMORHIJTBBEARHEVERHISYENIHREWARKACLEDIHAPPEARASYEASDUSEJPEYTE
DRETURSTNERIYNEHNENADCROUFNTCAYKMROWNIHTRAVELHNADSOGCEYOW
EALOYALLEFESDASDITGAHPOPULARLBCELIEVEDGNATEVERTNEOLDMOLKWIF
NTHABTNATTNENILLATCAFESDGAHMULLOMTUSSELHHTUMMEDGITNTREAHUR
EASDIMTNATGAHSOTESOUFNMORMAWETNEREGAHALHONIHPROLOSFEDVIFOU
RTOWARVELATTIWEGOREOSCUTITHEEWEDTONAVELITTLEEMMEYTOSWRCAFFI
SHATSISETBNEGAHWUYNTNEHAWEAHATMIMTBATSISETBSISETNEBCEFASTOYA
LLNIWGELLPREHERVEDCUTUSYNASFE

EASIER TO BREAK: Ciphertext 1 ($\chi^2 = 10.0$)

HARDER TO BREAK: Ciphertext 2 ($\chi^2 = 17.9$)

Why?

- The ciphertext with the **lower χ^2** after hill-climbing fits English letter frequencies better.

- A lower score usually means the cipher is **shorter** or **uses a key that preserves more of the original distribution** (e.g., a key that maps high-frequency letters to high-frequency letters).

- Very short texts (< 80 letters) give noisy frequency tables → higher χ^2 → harder to break automatically.